

Global Quality Required Tool GSOP-104-3-TL-5		
Version Number	Artificial Intelligence for Deviations	Associated Procedure
1		GSOP-104-3
Superseded Document		Page 1 of 4
None		

Purpose

This required tool describes the process of using Artificial Intelligence (AI) for deviations. This process includes tools to:

- Review and provide feedback on deviation description and immediate actions,
- categorize root cause(s),
- review and provide feedback on investigation details, and
- generate executive summaries for deviation reports.

Scope

In Scope:

Quality Deviations and Lab Investigations are in scope. The Virtual Mentor Tools, Root Cause Categorization Tool, and Executive Summary Generator Tool are available for use for Quality deviation records. Only the Root Cause Categorization Tool is available for use for laboratory investigations.

Out of Scope:

External deviation records where no further investigation is required are out of scope for the Root Cause Categorization, Virtual Mentor for Investigation, and Executive Summary Generator Tools. Other quality record types which are not deviation records are also out of scope. Private records are not accessible by the AI tools.

Table of Contents

1.0 Overview of Artificial Intelligence for Deviations	2
2.0 Virtual Mentor for Fact Finding	2
3.0 Root Cause Categorization	2
4.0 Virtual Mentor for Investigation.....	3
5.0 Executive Summary Generator (ESG)	3
Revision History	4

Global Quality Required Tool GSOP-104-3-TL-5		
Version Number	Artificial Intelligence for Deviations	Associated Procedure
1		GSOP-104-3
Superseded Document		Page 2 of 4
None		

1.0 Overview of Artificial Intelligence for Deviations

Artificial Intelligence (AI) tools aid in the execution of deviation record investigation and documentation. Use of the Root Cause Generator Tool is required. The remainder of the AI Tools are recommended to be used throughout the deviation process, but are not required. For all information and recommendations generated from the tools, the user must review the results/ output for accuracy and, if needed, make necessary changes to the information prior to finalizing the information for the deviation record.

2.0 Virtual Mentor for Fact Finding

Responsible for this Section: Fact Finder

Step	Activity	Related Information
1	Use Virtual Mentor for Fact Finding	The Virtual Mentor for Fact Finding is a tool which analyzes the content of the description and immediate steps to ensure that the content is complete and contains appropriate information in the fields. Use of the tool is not mandatory but highly encouraged to gain feedback prior to sending for READY Assessment. Refer to GSOP-104-1, Record Creation and Fact Finding .
2	Launch the Virtual Mentor	Launch the Virtual Mentor from the Action Menu to open the Fact Finder analysis. Information from the description and immediate steps fields automatically populate into the Virtual Mentor and the analysis will start.
3	Review the Recommendation	Review the recommendation(s) where provided to improve the description and/or immediate steps fields from the Virtual Mentor as the “human in the loop”. The human must decide whether to take action on feedback. Update the recommended text in the Global Deviation Database (GDD) if needed. Note: The Virtual Mentor for Fact Finding can be run multiple times on a record.

3.0 Root Cause Categorization

Responsible for this Section: Lead Investigator

Step	Activity	Related Information
1	Use Root Cause Categorization Tool	Use the Root Cause Categorization AI tool upon completion of the Investigation Details and Root Cause Analysis (RCA), if any, of a deviation record for the AI tool to provide suggested Root Cause(s) based on the Investigation Details. Use of the tool is required. Refer to GSOP-104-3, Investigation .
2	Launch the Tool	Launch the Root Cause Categorization tool when the deviation details have been completed, including details of the Root Cause Analysis performed. The deviation details will automatically populate in the Root Cause Categorization and the analysis will run.
3	Review Results	Once the AI Root Cause Categorization tool has completed the analysis of the deviation details, review the results as the “human in the loop” to determine which recommended root cause(s), if any, are appropriate to be added to the deviation record.
4	Add, Edit, or Delete Recommended Root Cause(s)	Condition: AI suggested root cause(s) needs to be changed Modify the recommended root cause results, as needed. If a different root cause is needed, one may be added manually or an existing result may be edited. If an incorrect result is included, it may be deleted.

Global Quality Required Tool GSOP-104-3-TL-5		
Version Number	Artificial Intelligence for Deviations	Associated Procedure
1		GSOP-104-3
Superseded Document		Page 3 of 4
None		

3.0 Root Cause Categorization

Responsible for this Section: Lead Investigator

Step	Activity	Related Information
5	Complete Root Cause Categorization	Once the appropriate Root Cause(s) are present, the categorization is completed and any root cause(s) which were not deleted within the tool will automatically be transferred to the deviation record in the Root Cause Object.

4.0 Virtual Mentor for Investigation

Responsible for this Section: Lead Investigator

Step	Activity	Related Information
1	Use Virtual Mentor for Investigation	The Virtual Mentor used at Investigation analyzes the content of the description, immediate steps, Quality Risk Evaluation, Investigation Details, CAPA Plan, Recurrence Check Details, and Effectiveness Check Plan to ensure that the content is complete and contains expected information in the fields. Use of the tool is not mandatory but highly encouraged to gain feedback prior to sending for Review/Approval. See GSOP-104-3, Investigation .
2	Launch the Virtual Mentor	Launch the Virtual Mentor from the Action Menu to open the Investigation analysis. The information from the deviation record fields will automatically populate into the Virtual Mentor and the analysis will run.
3	Review the Results	Review the output of the analysis performed on the deviation record fields. Review the recommendation(s) to improve the content from the Virtual Mentor as the “human in the loop”. The human must decide whether to take action on the feedback and will update the recommended text in the Global Deviation Database (GDD) as needed. Note: The Virtual Mentor for Investigation can be run multiple times on a record.

5.0 Executive Summary Generator (ESG)

Responsible for this Section: Lead Investigator

Step	Activity	Related Information
1	Use the ESG	The Executive Summary Generator (ESG) is available for use after the Virtual Mentor for Investigation is executed. The tool will utilize the fields on the deviation details and summary to generate a draft Executive Summary for the deviation investigation. Refer to GSOP-104-3, Investigation .
2	Launch the ESG	Launch the Virtual Mentor from the Action Menu to open the ESG. The information from the deviation record fields will automatically populate into the Virtual Mentor and the generator will start the analysis.
3	Review the Output	Review the executive summary as the “human in the loop”. The human will determine if the AI suggestion is appropriate prior to adding to deviation record. Note: the executive summary can be edited, as needed, in the GDD after sending it to the deviation record.
4	Send Draft Executive Summary to	Select “Send to Deviation Record” for the executive summary to automatically populate the summary into the Executive Summary field on the investigation details and summary section on the record.

Global Quality Required Tool GSOP-104-3-TL-5		
Version Number	Artificial Intelligence for Deviations	Associated Procedure
1		GSOP-104-3
Superseded Document		Page 4 of 4
None		

5.0 Executive Summary Generator (ESG)**Responsible for this Section:** Lead Investigator

Step	Activity	Related Information
	Deviation Report	
5	Changes to the Deviation Record After Executive Summary Generation	Edit the executive summary within the deviation record in the GDD, as needed.

Revision History

Version Number	Effective Date	List of Major Changes
1	Effective date viewable in eDMS.	First version of this global quality required tool.

Document Approval Signatures

Approved Date: 16 May 2025

Approval Verdict: Approve	Ciara Keohane (C206691@lilly.com) Approver 16-May-2025 14:14:58 GMT+0000
Approval Verdict: Approve	Bryce Hufford (A113383@lilly.com) Approver 16-May-2025 15:26:39 GMT+0000
Reviewer / Approver Additional Details	N/A